

1. ArrayIndexOutOfBoundsException

Question: Demonstrate ArrayIndexOutOfBoundsException in Java. **Steps for Execution:**

1. Create an array with 3 elements.
2. Try to access index 5.
3. Catch the exception and print a message.

Algorithm:

- Start program
- Initialize array
- Access invalid index
- Catch exception → Print message

UML Class Diagram:

```
+-----+
| ArrayIndexExample |
+-----+
| main(String[] args) |
```

Code:

```
public class ArrayIndexExample {
    public static void main(String[] args) {
        try {
            int[] arr = {1,2,3};
            System.out.println(arr[5]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Array index error!");
        }
    }
}
```

Output Screenshot (text):

Array index error!

2. NullPointerException

Question: Demonstrate NullPointerException in Java.

Steps: Assign null to a String and call .length().

Algorithm: Start → Assign null → Call method → Catch exception.

UML:

```
+-----+
| NullPointerException |
+-----+

+-----+
| main(String[] args) |
+-----+
```

Code:

```
public class NullPointerException {
    public static void main(String[] args) {
        try {
            String s = null;
            System.out.println(s.length());
        } catch (NullPointerException e) {
            System.out.println("Null pointer!");
        }
    }
}
```

Output:

Null pointer!

3. NumberFormatException

Question: Demonstrate NumberFormatException.

Steps: Parse invalid string to integer.

Algorithm: Start → Parse string → Catch exception.

UML:

```
+-----+
| NumberFormatExample |
+-----+
| main(String[] args) |
+-----+
```

Code:

```
public class NumberFormatExample {
    public static void main(String[] args) {
        try {
            int n = Integer.parseInt("abc");
        } catch (NumberFormatException e) {
            System.out.println("Number format error!");
        }
    }
}
```

Output:

Number format error!

4. ClassCastException

Question: Demonstrate ClassCastException.

Steps: Cast String object to Integer.

Algorithm: Start → Wrong cast → Catch exception.

UML:

```
+-----+
| ClassCastExample |
+-----+
| main(String[] args) |
+-----+
```

Code:

```
public class ClassCastExample {
    public static void main(String[] args) {
        try {
            Object o = "Hi";
            Integer i = (Integer) o;
        } catch (ClassCastException e) {
            System.out.println("Class cast error!");
        }
    }
}
```

Output:

Class cast error!

5. StringIndexOutOfBoundsException

Question: Demonstrate StringIndexOutOfBoundsException.

Steps: Access invalid char index.

Algorithm: Start → Create string → Access invalid index → Catch exception.

UML:

```

+-----+
| StringIndexExample |
+-----+

| main(String[] args) |

```

Code:

```

public class StringIndexExample {
    public static void main(String[] args) {
        try {
            String s = "Java";
            System.out.println(s.charAt(10));
        } catch (StringIndexOutOfBoundsException e) {
            System.out.println("String index error!");
        }
    }
}

```

Output:

String index error!

6. FileNotFoundException

Question: Demonstrate FileNotFoundException.

Steps: Try to open missing file → Catch exception.

Algorithm: Start → Open file → Catch exception.

UML:

```

+-----+
| FileNotFoundExample |
+-----+

| main(String[] args) |

```

CODE:

```
import java.io.*;

public class FileNotFoundExample {

    public static void main(String[] args) {

        try {

            FileReader fr = new FileReader("nofile.txt");

        } catch (FileNotFoundException e) {

            System.out.println("File not found!");

        }

    }

}
```

Output:

File not found!

7. IOException

Question: Demonstrate IOException.

Steps: Write to file → Catch exception.

Algorithm: Start → Write file → Catch exception.

UML:

```
+-----+
| IOExceptionExample |
+-----+

| main(String[] args) |
+-----+
```

Code:

```
import java.io.*;

public class IOExceptionExample {
```

```

public static void main(String[] args) {
    try {
        FileWriter fw = new FileWriter("test.txt");
        fw.write("Hello");
        fw.close();
    } catch (IOException e) {
        System.out.println("IO error!");
    }
}
}

```

Output:

IO error!

8. IllegalArgumentException

Question: Demonstrate IllegalArgumentException. Steps: Set invalid thread priority → Catch exception. Algorithm: Start → Create thread → Set invalid priority → Catch exception.

UML:

```

+-----+
| IllegalArgumentException |
+-----+
| main(String[] args)   |
+-----+

```

Code:

```

public class IllegalArgumentExample {
    public static void main(String[] args) {
        try {
            Thread t = new Thread();
            t.setPriority(20);

```

```

    } catch (IllegalArgumentException e) {
        System.out.println("Illegal argument!");
    }
}
}

```

Output:

Illegal argument!

9. IllegalStateException

Question: Demonstrate IllegalStateException. **Steps:** Use iterator remove wrongly → Catch exception. **Algorithm:** Start → Create iterator → Call remove → Catch exception.

UML:

```

+-----+
| IllegalStateException |
+-----+
| main(String[] args) |
+-----+

```

Code:

```

import java.util.*;

public class IllegalStateException {

    public static void main(String[] args) {

        try {

            Iterator<Integer> it = Arrays.asList(1,2,3).iterator();

            it.remove();

        } catch (IllegalStateException e) {

            System.out.println("Illegal state!");

        }

    }

}

```



```
}
```

Output:

Illegal state!

10. InterruptedException

Question: Demonstrate InterruptedException. Steps: Sleep thread → Catch exception.

Algorithm: Start → Sleep → Catch exception.

UML:

```
+-----+
| InterruptedException |
+-----+
| main(String[] args) |
+-----+
```

```
public class InterruptedException {
    public static void main(String[] args) {
        try {
            Thread.sleep(1000);
        } catch (InterruptedException e) {
            System.out.println("Thread interrupted!");
        }
    }
}
```

Output:

Thread interrupted!